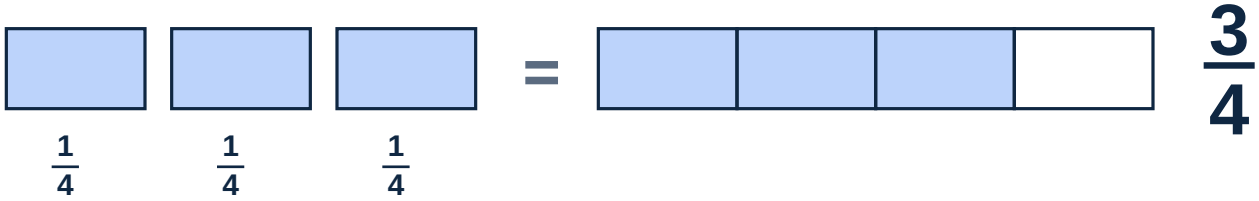
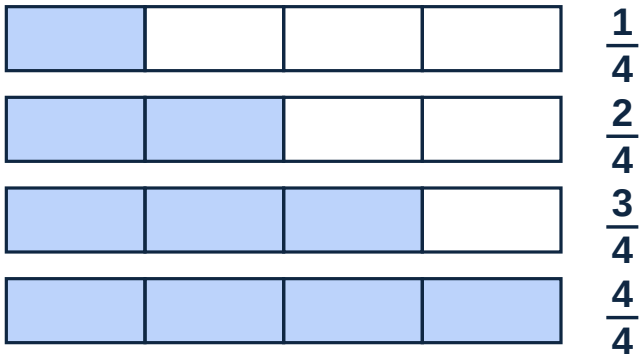
Counting Fractions

Count the pieces. 3 pieces of 1/4 is 3/4.

Put unit fractions together



Count by fourths



Top number: how many pieces.

Bottom number: how many equal parts make the whole.

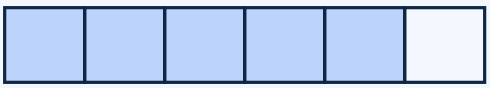
4 pieces of 1/4 fill the bar: that's the whole.

Say it in words

$\frac{2}{3}$ two thirds

$\frac{5}{8}$ five eighths

Your turn!



$\frac{5}{6}$

For teachers

Goal: a/b is a pieces of size 1/b (3.NF.1). Point at each piece and count: “one fourth, two fourths, three fourths.” The top number counts pieces; the bottom number names their size.

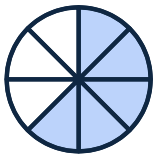
Counting Fractions: Practice

Top: how many pieces. Bottom: how many equal parts in the whole.

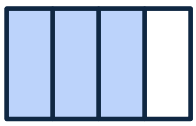
Part 1 Count the pieces How many pieces are shaded? What size is each one?

1) 

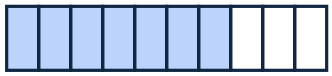
3 pieces of $\frac{1}{5}$ = $\frac{3}{5}$

2) 

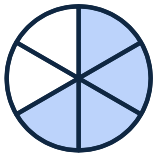
5 pieces of $\frac{1}{8}$ = $\frac{5}{8}$

3) 

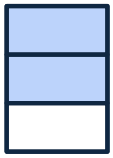
3 pieces of $\frac{1}{4}$ = $\frac{3}{4}$

4) 

7 pieces of $\frac{1}{10}$ = $\frac{7}{10}$

5) 

4 pieces of $\frac{1}{6}$ = $\frac{4}{6}$

6) 

2 pieces of $\frac{1}{3}$ = $\frac{2}{3}$

7) 

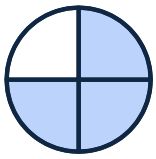
6 pieces of $\frac{1}{8}$ = $\frac{6}{8}$

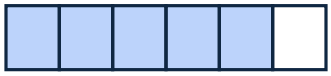
8) 

2 pieces of $\frac{1}{5}$ = $\frac{2}{5}$

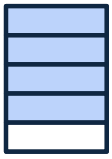
Counting Fractions: Practice (continued)

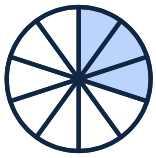
Part 2 Name the fraction

9)  $\frac{3}{4}$

10)  $\frac{5}{6}$

11)  $\frac{3}{8}$

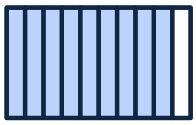
12)  $\frac{4}{5}$

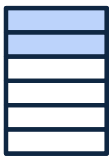
13)  $\frac{3}{10}$

14)  $\frac{4}{6}$

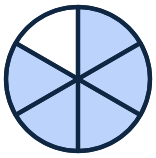
15)  $\frac{2}{3}$

16)  $\frac{2}{5}$

17)  $\frac{9}{10}$

18)  $\frac{2}{6}$

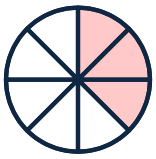
19)  $\frac{7}{8}$

20)  $\frac{5}{6}$

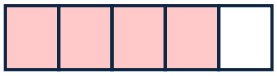
Counting Fractions: Practice (continued)

Part 3 Shade it

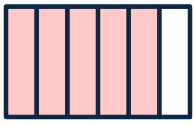
21) Shade $\frac{3}{8}$



22) Shade $\frac{4}{5}$



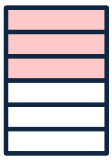
23) Shade $\frac{5}{6}$



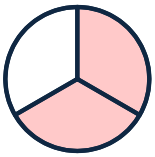
24) Shade $\frac{2}{4}$



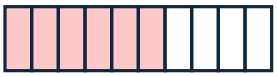
25) Shade $\frac{3}{6}$



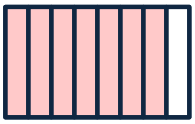
26) Shade $\frac{2}{3}$



27) Shade $\frac{6}{10}$



28) Shade $\frac{7}{8}$



Part 4 Count by fractions No help this time. Fill in the missing fractions.

29) $\frac{1}{5}$ $\frac{2}{5}$ $\frac{\boxed{3}}{5}$ $\frac{\boxed{4}}{5}$ $\frac{5}{5}$

30) $\frac{1}{8}$ $\frac{2}{8}$ $\frac{3}{8}$ $\frac{\boxed{4}}{8}$ $\frac{\boxed{5}}{8}$

31) $\frac{2}{6}$ $\frac{3}{6}$ $\frac{\boxed{4}}{6}$ $\frac{\boxed{5}}{6}$ $\frac{6}{6}$

32) $\frac{1}{3}$ $\frac{\boxed{2}}{3}$ $\frac{3}{3}$

Counting Fractions: Practice (continued)

Part 4 Count by fractions (continued)

33) $\frac{4}{10}$ $\frac{5}{10}$ $\frac{\boxed{6}}{10}$ $\frac{\boxed{7}}{10}$ $\frac{8}{10}$

34) $\frac{\boxed{1}}{4}$ $\frac{2}{4}$ $\frac{3}{4}$ $\frac{\boxed{4}}{4}$

35) $\frac{3}{8}$ $\frac{4}{8}$ $\frac{5}{8}$ $\frac{\boxed{6}}{8}$ $\frac{\boxed{7}}{8}$

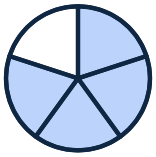
36) $\frac{1}{6}$ $\frac{\boxed{2}}{6}$ $\frac{\boxed{3}}{6}$ $\frac{4}{6}$

Part 5 Shaded and not shaded Together they make the whole.

37) 

shaded $\frac{\boxed{3}}{8}$

not shaded $\frac{\boxed{5}}{8}$

38) 

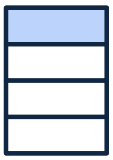
shaded $\frac{\boxed{4}}{5}$

not shaded $\frac{\boxed{1}}{5}$

39) 

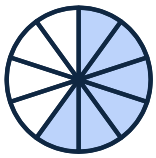
shaded $\frac{\boxed{1}}{6}$

not shaded $\frac{\boxed{5}}{6}$

40) 

shaded $\frac{\boxed{1}}{4}$

not shaded $\frac{\boxed{3}}{4}$

41) 

shaded $\frac{\boxed{6}}{10}$

not shaded $\frac{\boxed{4}}{10}$

42) 

shaded $\frac{\boxed{2}}{6}$

not shaded $\frac{\boxed{4}}{6}$

Counting Fractions: Practice (continued)

Part 6 Pieces to fractions No help this time.

43) 4 pieces of $\frac{1}{6} = \frac{4}{6}$

44) 2 pieces of $\frac{1}{3} = \frac{2}{3}$

45) 7 pieces of $\frac{1}{8} = \frac{7}{8}$

46) 3 pieces of $\frac{1}{5} = \frac{3}{5}$

47) 6 pieces of $\frac{1}{10} = \frac{6}{10}$

48) 2 pieces of $\frac{1}{4} = \frac{2}{4}$

49) 5 pieces of $\frac{1}{8} = \frac{5}{8}$

50) 9 pieces of $\frac{1}{10} = \frac{9}{10}$

51) 3 pieces of $\frac{1}{6} = \frac{3}{6}$

52) 4 pieces of $\frac{1}{5} = \frac{4}{5}$

53) 2 pieces of $\frac{1}{8} = \frac{2}{8}$

54) 5 pieces of $\frac{1}{10} = \frac{5}{10}$

Counting Fractions: Practice (continued)

Part 7 Fractions to pieces No help this time.

55) $\frac{3}{8} = \boxed{3}$ pieces of $\frac{1}{8}$

56) $\frac{5}{6} = \boxed{5}$ pieces of $\frac{1}{6}$

57) $\frac{2}{5} = \boxed{2}$ pieces of $\frac{1}{5}$

58) $\frac{3}{10} = \boxed{3}$ pieces of $\frac{1}{10}$

59) $\frac{6}{8} = \boxed{6}$ pieces of $\frac{1}{8}$

60) $\frac{2}{6} = \boxed{2}$ pieces of $\frac{1}{6}$

61) $\frac{3}{4} = \boxed{3}$ pieces of $\frac{1}{4}$

62) $\frac{8}{10} = \boxed{8}$ pieces of $\frac{1}{10}$

63) $\frac{4}{8} = \boxed{4}$ pieces of $\frac{1}{8}$

64) $\frac{2}{10} = \boxed{2}$ pieces of $\frac{1}{10}$

65) $\frac{4}{10} = \boxed{4}$ pieces of $\frac{1}{10}$

66) $\frac{7}{10} = \boxed{7}$ pieces of $\frac{1}{10}$

Counting Fractions: Practice (continued)

Part 8 Words and numbers No help this time.

67) two fifths = $\frac{2}{5}$

68) five sixths = $\frac{5}{6}$

69) three eighths = $\frac{3}{8}$

70) seven tenths = $\frac{7}{10}$

71) three fourths = $\frac{3}{4}$

72) four sixths = $\frac{4}{6}$

73) two thirds = $\frac{2}{3}$

74) five eighths = $\frac{5}{8}$

75) six tenths = $\frac{6}{10}$

76) four fifths = $\frac{4}{5}$

Counting Fractions: Practice (continued)

Part 9 How much is left? No help this time. Shaded and not shaded make the whole.

77) $\frac{3}{8}$ shaded, $\frac{5}{8}$ not shaded

78) $\frac{1}{6}$ shaded, $\frac{5}{6}$ not shaded

79) $\frac{4}{10}$ shaded, $\frac{6}{10}$ not shaded

80) $\frac{2}{5}$ shaded, $\frac{3}{5}$ not shaded

81) $\frac{1}{4}$ shaded, $\frac{3}{4}$ not shaded

82) $\frac{5}{8}$ shaded, $\frac{3}{8}$ not shaded

83) $\frac{2}{3}$ shaded, $\frac{1}{3}$ not shaded

84) $\frac{3}{6}$ shaded, $\frac{3}{6}$ not shaded

Part 10 How many pieces?

85) $\frac{1}{3} = 1$ pieces of $\frac{1}{3}$

86) $\frac{9}{12} = 9$ pieces of $\frac{1}{12}$

87) $\frac{1}{10} = 1$ pieces of $\frac{1}{10}$

88) $\frac{4}{5} = 4$ pieces of $\frac{1}{5}$

Counting Fractions: Practice (continued)

Part 10 How many pieces? (continued)

89) $\frac{1}{5} = \boxed{1}$ pieces of $\frac{1}{5}$

90) $\frac{3}{5} = \boxed{3}$ pieces of $\frac{1}{5}$

91) $\frac{1}{4} = \boxed{1}$ pieces of $\frac{1}{4}$

92) $\frac{5}{12} = \boxed{5}$ pieces of $\frac{1}{12}$

93) $\frac{1}{6} = \boxed{1}$ pieces of $\frac{1}{6}$

94) $\frac{4}{12} = \boxed{4}$ pieces of $\frac{1}{12}$

95) $\frac{1}{2} = \boxed{1}$ pieces of $\frac{1}{2}$

96) $\frac{2}{12} = \boxed{2}$ pieces of $\frac{1}{12}$

97) $\frac{1}{8} = \boxed{1}$ pieces of $\frac{1}{8}$

98) $\frac{7}{8} = \boxed{7}$ pieces of $\frac{1}{8}$

99) $\frac{8}{12} = \boxed{8}$ pieces of $\frac{1}{12}$

100) $\frac{11}{12} = \boxed{11}$ pieces of $\frac{1}{12}$

Counting Fractions: Practice (continued)

Part 10 How many pieces? (continued)

101) $\frac{7}{12} = \boxed{7}$ pieces of $\frac{1}{12}$

102) $\frac{1}{12} = \boxed{1}$ pieces of $\frac{1}{12}$

103) $\frac{3}{12} = \boxed{3}$ pieces of $\frac{1}{12}$

104) $\frac{6}{10} = \boxed{6}$ pieces of $\frac{1}{10}$

105) $\frac{3}{6} = \boxed{3}$ pieces of $\frac{1}{6}$

106) $\frac{9}{10} = \boxed{9}$ pieces of $\frac{1}{10}$

107) $\frac{5}{10} = \boxed{5}$ pieces of $\frac{1}{10}$

108) $\frac{10}{12} = \boxed{10}$ pieces of $\frac{1}{12}$

Part 11 Shaded and not shaded

109) $\frac{2}{8}$ shaded, $\frac{\boxed{6}}{8}$ not shaded

110) $\frac{11}{12}$ shaded, $\frac{\boxed{1}}{12}$ not shaded

111) $\frac{2}{4}$ shaded, $\frac{\boxed{2}}{4}$ not shaded

112) $\frac{1}{2}$ shaded, $\frac{\boxed{1}}{2}$ not shaded

Counting Fractions: Practice (continued)

Part 11 Shaded and not shaded (continued)

113) $\frac{2}{6}$ shaded, $\frac{4}{6}$ not shaded

114) $\frac{1}{3}$ shaded, $\frac{2}{3}$ not shaded

115) $\frac{2}{10}$ shaded, $\frac{8}{10}$ not shaded

116) $\frac{9}{12}$ shaded, $\frac{3}{12}$ not shaded

117) $\frac{7}{8}$ shaded, $\frac{1}{8}$ not shaded

118) $\frac{1}{10}$ shaded, $\frac{9}{10}$ not shaded

119) $\frac{8}{12}$ shaded, $\frac{4}{12}$ not shaded

120) $\frac{3}{5}$ shaded, $\frac{2}{5}$ not shaded

121) $\frac{4}{12}$ shaded, $\frac{8}{12}$ not shaded

122) $\frac{6}{10}$ shaded, $\frac{4}{10}$ not shaded

123) $\frac{1}{5}$ shaded, $\frac{4}{5}$ not shaded

124) $\frac{6}{8}$ shaded, $\frac{2}{8}$ not shaded

Counting Fractions: Practice (continued)

Part 11 Shaded and not shaded (continued)

125) $\frac{4}{6}$ shaded, $\frac{2}{6}$ not shaded

126) $\frac{5}{6}$ shaded, $\frac{1}{6}$ not shaded

Part 12 Remember? Problems from earlier packets.

127) 8 groups of 3 = 24

128) 6 cm by 8 cm: area = 48 sq cm

129) Round 547 to the nearest ten: 550

130) 4 groups of 7 = 28

131) Round 351 to the nearest hundred: 400

132) Round 762 to the nearest ten: 760

Part 13 Word problems Draw a picture.

133) A pizza has 8 equal slices. Tom eats 3 slices. What fraction of the pizza does he eat?

Number sentence 3 pieces of 1/8 = 3/8

Answer 3/8 of the pizza

Draw or show your work

134) A garden has 5 equal rows. 2 rows are beans. What fraction of the garden is beans?

Number sentence 2 pieces of 1/5 = 2/5

Answer 2/5 of the garden

Draw or show your work

135) A chocolate bar has 6 equal pieces. Ben eats 1 piece. What fraction of the bar is left?

Number sentence 1/6 eaten, 5/6 left

Answer 5/6 of the bar

Draw or show your work

Counting Fractions: Practice (continued)

Part 13 Word problems (continued)

136) Mia paints 4 of the 10 equal parts of a fence. What fraction is not painted yet?

Number sentence 4/10 painted, 6/10 not painted

Answer 6/10 of the fence

Draw or show your work

137) Challenge: Sam eats 2/8 of a pie. Kim eats 3/8 of it. How much of the pie do they eat in all?

Number sentence 2 eighths and 3 eighths = 5 eighths

Answer 5/8 of the pie

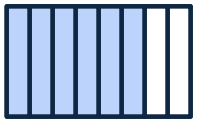
Draw or show your work

Check-In: Counting Fractions

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 Name the fraction

1)  $\frac{6}{8}$

2)  $\frac{7}{10}$

Part 2 Shade it

3) Shade $\frac{2}{5}$ 

4) Shade $\frac{4}{8}$ 

Part 3 Count by fractions

5) $\frac{1}{5}$ $\frac{2}{5}$ $\frac{3}{5}$ $\frac{4}{5}$

6) $\frac{2}{10}$ $\frac{3}{10}$ $\frac{4}{10}$ $\frac{5}{10}$

Part 4 Pieces and fractions

7) $\frac{2}{4} = 2$ pieces of $\frac{1}{4}$

8) $\frac{4}{6} = 4$ pieces of $\frac{1}{6}$

Part 5 How much is left?

9) $\frac{1}{8}$ shaded, $\frac{7}{8}$ not shaded

10) $\frac{9}{10}$ shaded, $\frac{1}{10}$ not shaded